APPLIED AI & ROBOTICS

PORTFOLIO PITCH DECK

Showcasing scalable machine learning solutions, foundational algorithm design, and generative AI architectures developed through rigorous academic and practical implementation.

ENGINEER	INSTITUTION	PROGRAM
Javon Darby	Houston City College	Deep Learning (ITAI 2376)

[PRIMARY REPOSITORY: GITHUB]

EXECUTIVE PROFILE & TECHNICAL CAPABILITIES

THE PROFILE

Dedicated Applied AI and Robotics engineer focused on bridging the gap between theoretical mathematics and framework-driven production systems.

Proven track record of designing, training, and deploying complex neural network architectures, handling everything from data preprocessing pipelines to optimal hyperparameter tuning.

THE ARSENAL

- PyTorch & Torchvision
- TensorFlow & Keras
- Gymnasium (RL)
- Hugging Face (Transformers)
- Pandas & NumPy
- Computer Vision (CNNs)
- Sequence Models (LSTMs)
- GenAI / Diffusion

05
Complete ML Projects

10K+
Lines of Production Code

100%
Functional Repository

SYS.01 // GENERATIVE DIFFUSION ARCHITECTURE

PROBLEM STATEMENT

To demystify state-of-the-art Generative AI systems (like Midjourney/DALL-E) by engineering a probabilistic diffusion model entirely from scratch.

THE ARCHITECTURE

- Deep **Convolutional U-Net** backbone built in PyTorch.
- Integrated **Sinusoidal Positional Embeddings** for precise timestep conditioning.
- Mathematical implementation of **Markov chains** for iterative noise injection and denoising.

EXECUTION & RESULTS

High Fidelity
Output Resolution

The model demonstrated a profound understanding of the latent space distribution. It successfully learned to reverse entropy, generating clear, recognizable digits from pure static Gaussian noise. This project proves an advanced capability in handling complex matrix operations, GPU acceleration, and probabilistic programming.

SYS.02 // AUTONOMOUS CONTROL & SEQUENCE MODELS

REINFORCEMENT LEARNING AGENT

Trained an agent to balance a pole in the CartPole physics environment, demonstrating control theory application.

- Implemented the **Bellman Equation** for Q-Learning.
- Engineered an **Epsilon-Greedy policy** for the exploration-exploitation dilemma.
- **Result:** Agent survived maximum 500 steps.

NLP & TRANSFORMER ARCHITECTURE

Conducted an empirical comparison between classical Recurrent networks and modern Attention mechanisms.

- Built Bidirectional **LSTMs** and **GRUs** from scratch.
- Fine-tuned pre-trained **DistilBERT** LLMs using Hugging Face for text classification.
- **Result:** Demonstrated superiority of Self-Attention over sequential recurrence.

THE VALUE PROPOSITION

By bridging the gap between advanced academic theory and functional, deployable code, I deliver intelligent systems that solve complex computational problems. I am equipped to drive innovation in GenAI, Autonomous Systems, and Predictive Modeling.

Ready for Deployment

REVIEW THE CODE // GITHUB

CONNECT // LINKEDIN